CHAPTER EXERCISE: Answers for this chapter start on page 276.

A calculator is allowed on the following questions.

1

Reid wants to purchase a rug that has a price of \$150.00. He has a coupon that would reduce the cost of the rug by k%. If the coupon would reduce the cost of the rug by \$12.75, what is the value of k ?

2

In March, a city zoo attracted 32,000 visitors to its polar bear exhibit. In April, the number of visitors to the exhibit increased by 15%. How many visitors did the zoo attract to its polar bear exhibit in April?

A) 32,150
B) 32,480
C) 35,200
D) 36,800

3

Miguel is following a recipe for marinara sauce that requires half a tablespoon of vinegar. If one cup is equivalent to 16 tablespoons, approximately what percent of a cup of vinegar is the amount required by the recipe?

A) 2.3%
B) 3.1%
C) 9.4%
D) 12.5%

4

If x is 50% larger than z, and y is 20% larger than z, then x is what percent larger than y ?

A) 15%
B) 20%
C) 25%
D) 30%

5

Veronica has a bank account that earns m% interest compounded annually. If she opened the account with \$200, the expression $\$200(x)^t$ represents the amount in the account after t years. Which of the following gives x in terms of m ?

A) $1 + .01m$
B) $1 + m$
C) $1 - m$
D) $1 + 100m$

6

A charity organization collected 2,140 donations last month. With the help of 50 additional volunteers, the organization collected 2,690 donations this month. To the nearest tenth of a percent, what was the percent increase in the number of donations the charity organization collected?

A) 20.4%
B) 20.7%
C) 25.4%
D) 25.7%

7

The discount price of a book is 20% less than the retail price. James manages to purchase the book at 30% off the discount price at a special book sale. What percent of the retail price did James pay?

A) 42%

B) 48%

C) 50%

D) 56%

8

Each day, Robert eats 40% of the pistachios left in his jar at that time. At the end of the second day, 27 pistachios remain. How many pistachios were in the jar at the start of the first day?

A) 75

B) 80

C) 85

D) 95

9

Joanne bought a doll at a 10 percent discount off the original price of $105.82. However, she had to pay a sales tax of x% on the discounted price. If the total amount she paid for the doll was $100, what is the value of x ?

A) 2

B) 3

C) 4

D) 5

10

The number of dishes served by a restaurant during dinner was 17.5% greater than the number of dishes served during lunch. If the restaurant served 940 dishes during dinner, how many more dishes did the restaurant serve during dinner than during lunch?

11

In 2010, the number of houses built in Town A was 25 percent greater than the number of houses built in Town B. If 70 houses were built in Town A during 2010, how many were built in Town B?

12

Over a two week span, John ate 20 pounds of chicken wings and 15 pounds of hot dogs. Kyle ate 20 percent more chicken wings and 40 percent more hot dogs. Considering only chicken wings and hot dogs, Kyle ate approximately x percent more food, by weight, than John. What is x (rounded to the nearest percent)?

A) 25

B) 27

C) 29

D) 30

13

Jane is playing a board game in which she must collect as many cards as possible. On her first turn, she loses 18 percent of her cards. On the second turn, she increases her card count by 36 percent. If her final card count after these two turns is n, which of the following represents her starting card count in terms of n ?

A) $\frac{n}{(1.18)(0.64)}$

B) $(1.18)(0.64)n$

C) $\frac{n}{(1.36)(0.82)}$

D) $(0.82)(1.36)n$

14

Due to deforestation, researchers expect the deer population to decline by 6 percent every year. If the current deer population is 12,000, what is the approximate expected population size 10 years from now?

A) 4800

B) 6460

C) 7240

D) 7980

15

A small clothing store sells 3 different types of accessories: 20% are scarves, 60% are ties, and the other 40 accessories are belts. If half of the ties are replaced with scarves, how many scarves will the store have?

16

Omar currently holds a government bond that has a market value of \$900. Each year, the market value of the bond is expected to be 20% higher than its market value the year before. If the expression $900(1 + p)$, where p is a constant, represents the expected market value of the bond after 3 years, what is the value of p ?

17

Sims spent x dollars on groceries in 2015. She spent 34% more on groceries in 2016 than in 2015, and she spent 145% more on groceries in 2017 than in 2016. Which of the following expressions represents the amount, in dollars, Sims spent on groceries in 2017?

A) $(2.45)(0.34x)$

B) $(1.45)(0.34x)$

C) $(2.45)(1.34x)$

D) $(1.45)(1.34x)$

18

In 2016, County A and County B collected the same amount of taxes. In 2017, the amount of taxes collected by County A decreased by 25% and the amount of taxes collected by County B increased by 20%. If County A collected 60 million dollars of taxes in 2017, what was the amount of taxes, in millions of dollars, County B collected in 2017?

A) 54

B) 78

C) 90

D) 96

19

Daniel has \$1000 in a checking account and \$3000 in a savings account. The checking account earns him 1 percent interest compounded annually. The savings account earns him 6 percent interest compounded annually. Assuming he leaves both these accounts alone, which of the following represents how much more interest Daniel will have earned from the savings account than from the checking account after 5 years?

A) $3,000(1.06)^5 - 1,000(1.01)^5$

B) $3,000(1.06)(5) - 1,000(1.01)(5)$

C) $(3,000(1.06)^5 - 3,000) - (1,000(1.01)^5 - 1,000)$

D) $(3,000(1.06)(5) - 3,000) - (1,000(1.01)(5) - 1,000)$

20

$$P\left(1+\frac{r}{100}\right)^5$$

The expression above gives the population of leopards after five years during which an initial population of P leopards grew by r percent each year. Which of the following expressions gives the percent increase in the leopard population over these five years?

A) $\left(1+\frac{r}{100}\right)^5$

B) $\frac{\left(1+\frac{r}{100}\right)^5 - 1}{\left(1+\frac{r}{100}\right)^5} \times 100$

C) $\left[\left(1+\frac{r}{100}\right)^5 - 1\right] \times 100$

D) $\left(1+\frac{r}{100}\right)^5 \times 100$

Chapter 2: Percent

CHAPTER EXERCISE:

1. $\boxed{8.5}$ $\dfrac{12.75}{150} = 0.085 = 8.5\%$

2. $\boxed{D}$ $32,000(1.15) = 36,800$

3. $\boxed{B}$ $\dfrac{0.5}{16} = 0.03125 \approx 3.1\%$

4. $\boxed{C}$ Let $z = 100$. Then $x = 1.50(100) = 150$ and $y = 1.20(100) = 120$. x is

$$\frac{150 - 120}{120} = \frac{30}{120} = 25\%$$

larger than y.

5. $\boxed{A}$ Each year, Veronica keeps whatever she has in her account plus the interest on that amount. Because m is a percentage, we can convert it to a decimal by dividing it by 100, giving us $0.01m$. Therefore, $x = 1 + 0.01m$.

6. $\boxed{D}$ $\dfrac{\text{new value} - \text{old value}}{\text{old value}} \times 100\% = \dfrac{2,690 - 2,140}{2,140} \times 100\% \approx 25.7\%$

7. $\boxed{D}$ Let the original price of the book be \$100. Then James bought the book at $100(1 - 0.20)(1 - 0.30) = 100(0.80)(0.70) =$ \$56, which is $\dfrac{56}{100} = 56\%$ of the original price.

8. $\boxed{A}$ Let x be the number of pistachios at the start. At the end of each day, what's left is $1 - 0.40 = 0.60$ of the day's starting amount. Over two days,

$$\begin{aligned} x(0.60)(0.60) &= 27 \\ 0.36x &= 27 \\ x &= 75 \end{aligned}$$

9. $\boxed{D}$ Let x be the sales tax (as a decimal for now). We'll convert it to a percent at the end.

$$\begin{aligned} 105.82(.90)(1 + x) &= 100 \\ 1 + x &= \frac{100}{(105.82)(.90)} \\ x &= \frac{100}{(105.82)(.90)} - 1 \\ x &= 0.05 = 5\% \end{aligned}$$

10. $\boxed{140}$ Let x be the number of dishes served during lunch. Then

$$\begin{aligned} 1.175x &= 940 \\ x &= 800 \end{aligned}$$

Therefore, $940 - 800 = 140$ more dishes were served during dinner.

11. $\boxed{56}$

$$\begin{aligned} A &= (1.25)(B) \\ 70 &= (1.25)(B) \\ 56 &= B \end{aligned}$$

12. $\boxed{C}$ Kyle ate $20(1.20) = 24$ pounds of chicken wings and $15(1.40) = 21$ pounds of hot dogs. That's a total of $24 + 21 = 45$ pounds of food. John had $20 + 15 = 35$ pounds of food. The percent increase from John to Kyle is

$$\frac{45 - 35}{35} \approx .29 = 29\%$$

13. $\boxed{C}$ Let her starting card count be x. A loss of 18 percent reduces her total to $(0.82)x$. From there, an increase of 36 percent gets the total to $(1.36)(0.82)x$. Now,

$$\begin{aligned} (1.36)(0.82)x &= n \\ x &= \frac{n}{(1.36)(0.82)} \end{aligned}$$

14. [B] $12,000(0.94)^{10} \approx 6,460$.

15. [100] Since scarves and ties make up 80% of the accessories, the 40 belts must account for 20%. Letting the total number of accessories be x,

$$20\% \text{ of } x = 40$$
$$\frac{1}{5}x = 40$$
$$x = 200$$

There are 200 accessories in the store. Hopefully you're able to get this without having to make an equation, but there's no harm in a little algebra! Now we can determine that there are $\frac{1}{5} \times 200 = 40$ scarves and $\frac{3}{5} \times 200 = 120$ ties. Half of the 120 ties (60 ties) are replaced with scarves, so the store will end up with $40 + 60 = 100$ scarves.

16. [.728] After 3 years, the market value of the bond is $900(1.2)^3 = 900(1.728) = 900(1 + 0.728)$. Therefore, $p = .728$

17. [C] To get the final value after a percent increase, you have to multiply the initial value by 1 plus the percentage (as a decimal). So in 2016, Sims must have spent $1.34x$ dollars on groceries. In 2017, she must have spent $(1 + 1.45)(1.34x) = (2.45)(1.34x)$ dollars on groceries.

18. [D] Let x be the amount of taxes, in millions of dollars, collected by County A in 2016. Since the taxes decreased by 25% from 2016 to 2017,

$$(1 - 0.25)x = 60$$
$$0.75x = 60$$
$$x = 80$$

Because County B collected the same amount as County A in 2016, County B also collected 80 million dollars of taxes in 2016. In 2017, County B collected 20% more than in 2016, so County B must have collected $80(1.20) = 96$ million dollars in 2017.

19. [C] The total amount in the savings account after 5 years will be $3,000(1.06)^5$, but the interest earned will be $3,000(1.06)^5 - 3,000$. The total amount in the checking account after 5 years will be $1,000(1.01)^5$, but the interest earned will be $1,000(1.01)^5 - 1,000$. With a larger initial deposit and a higher interest rate, it's obvious the savings account will have earned more interest. The difference in earned interest will be $(3,000(1.06)^5 - 3,000) - (1,000(1.01)^5 - 1,000)$.

20. [C] The percent change is the new minus the old over the old times 100. Notice that the P's cancel out.

$$\frac{P\left(1 + \frac{r}{100}\right)^5 - P}{P} \times 100 =$$
$$\left[\left(1 + \frac{r}{100}\right)^5 - 1\right] \times 100$$